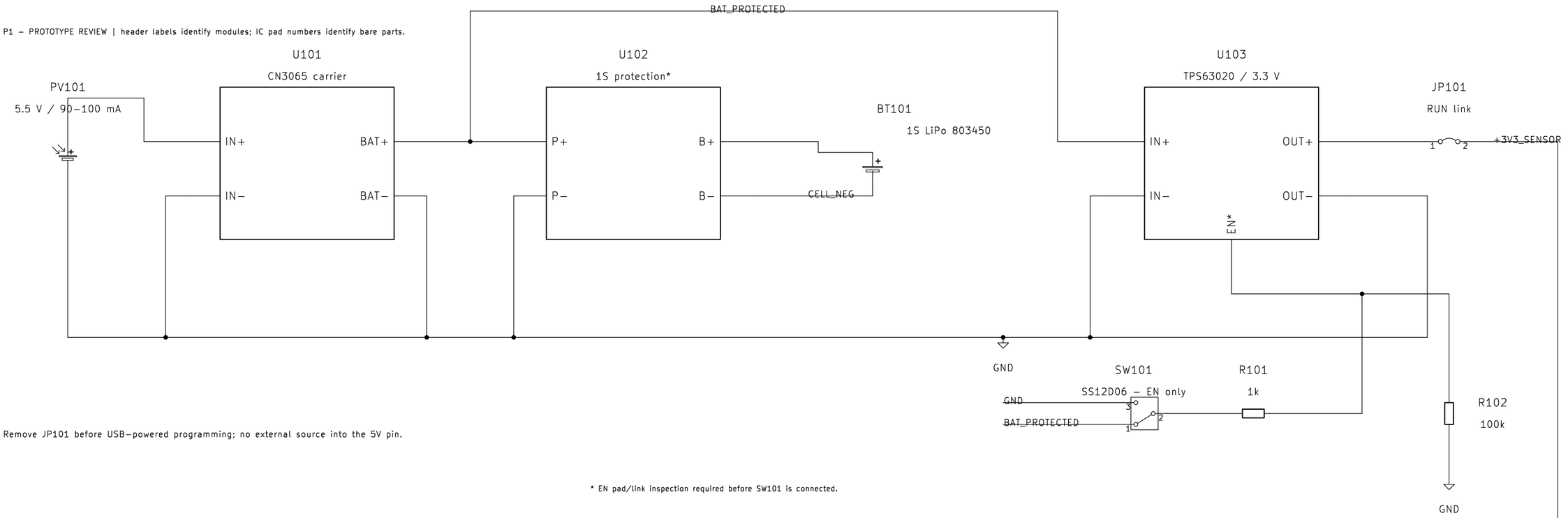
# PASADA | 01 SENSOR / SOLAR PUCK

P1 — PROTOTYPE REVIEW | header labels identify modules; IC pad numbers identify bare parts.




Remove JP101 before USB-powered programming; no external source into the 5V pin.

* EN pad/link inspection required before SW101 is connected.

1. Verify carrier pads, 3.30 V output and EN access. SS12D06 controls EN only.

2. Fit U102 only when needed for protection. Charger/load returns use P–; CELL_NEG stays separate.

3. CN3065: input <=6 V including panel Voc. Verify charge current, temperature protection and charging under load.

Manually routed. Not a field-installation release.

**PASADA / LEAR — engineering review**

Sheet: /
File: 01_sensor_puck.kicad_sch

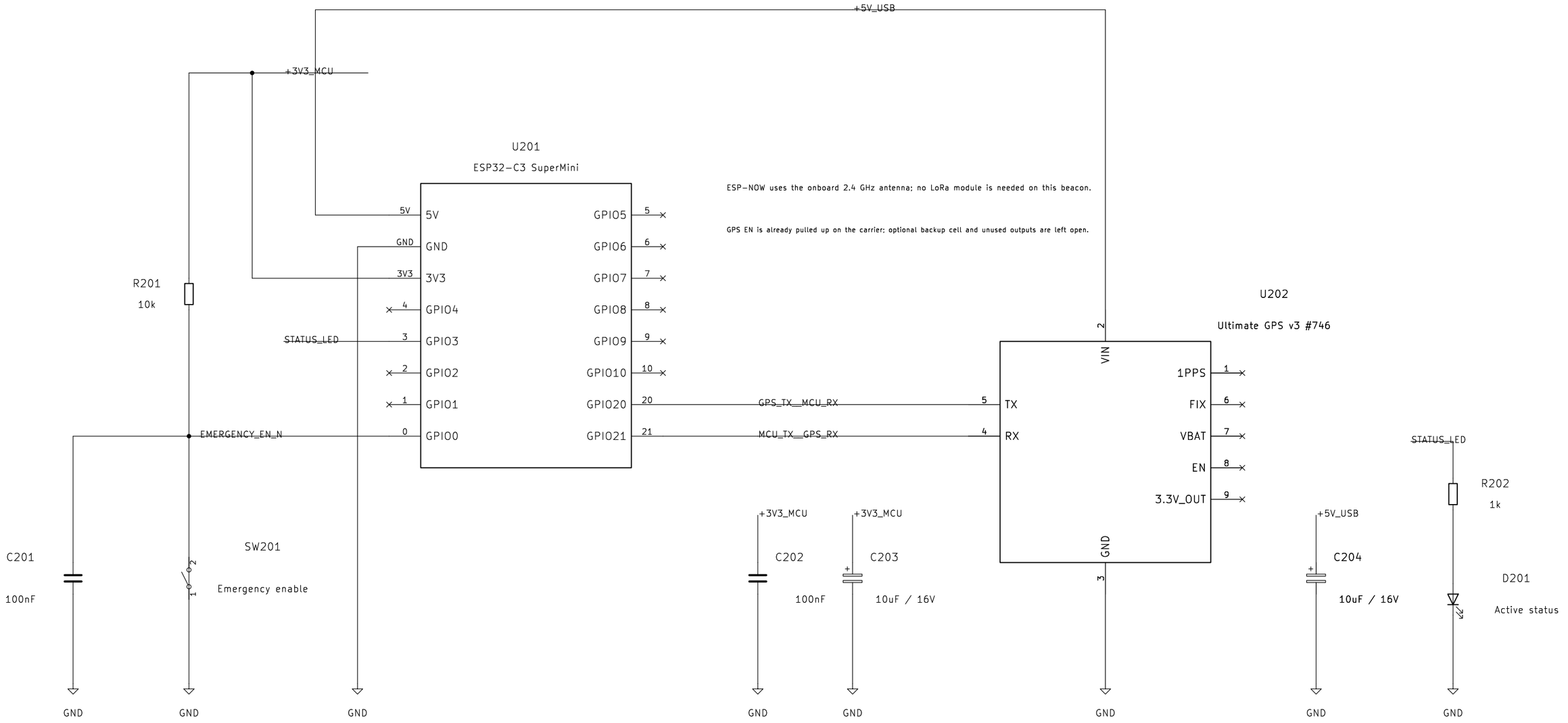
***Title: 01 SENSOR / SOLAR PUCK***

| Size: A3 | Date: 2026-10-31 | **Rev: P1** |
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| KiCad E.D.A. | kicad-cli 7.0.11+dfsg-1build4 | Id: 1/1 |

# PASADA | 02  EMERGENCY VEHICLE BEACON

P1 — PROTOTYPE REVIEW | header labels identify modules; IC pad numbers identify bare parts.

PROPOSED GPS: Adafruit Ultimate GPS v3 #746. Power through U201 onboard USB-C only (regulated 5 V USB).




1. GPS TX(5) -> GPIO20 RX; GPIO21 TX -> GPS RX(4). Use UART0 for GPS and native USB for logs.
2. SW201 is a firmware request. GPIO2/8/9 stay unloaded; signal safety remains with the coordinator.
3. Use regulated 5 V USB from a bench or protected vehicle adapter. Keep raw 12/24 V outside this circuit.

Manually routed. Not a field-installation release.
**PASADA / LEAR — engineering review**

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File: 02_emergency_beacon.kicad_sch

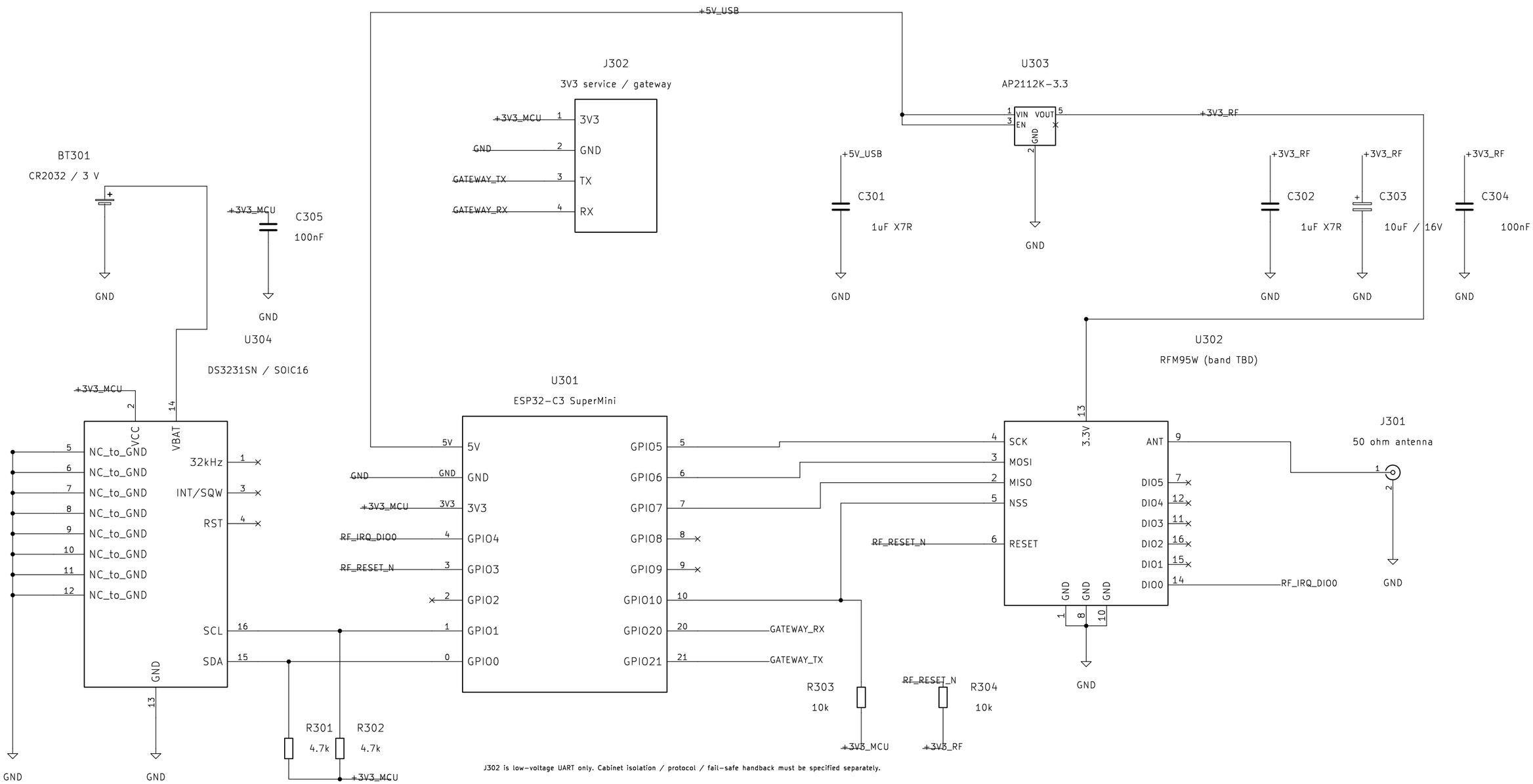
***Title: 02  EMERGENCY VEHICLE BEACON***

| Size: A3 | Date: 2026-10-31 | **Rev: P1** |
|---|---|---|
| KiCad E.D.A. | kicad-cli 7.0.11+dfsg-1build4 | Id: 1/1 |

# PASADA | 03 INTERSECTION COORDINATOR

P1 — PROTOTYPE REVIEW | header labels identify modules; IC pad numbers identify bare parts.

Power through U301 onboard USB-C only. Proposed DS3231SN + separate AP2112K-3.3 supply for RFM95W.




J302 is low-voltage UART only. Cabinet isolation / protocol / fail-safe handback must be specified separately.

1. I2C SDA/SCL=GPIO0/1. SPI SCK/MOSI/MISO/NSS=GPIO5/6/7/10. RF reset/IRQ=GPIO3/4.

2. DS3231SN pads 5–12 –> GND. CR2032 is not charged. MCU and RF regulator outputs remain separate.

3. Match RFM95W band and antenna. Use RF breakout/coax. Cabinet protocol and isolation remain to be specified.

Manually routed. Not a field-installation release.
**PASADA / LEAR — engineering review**

Sheet: /
File: 03_intersection_coordinator.kicad_sch

***Title: 03 INTERSECTION COORDINATOR***

| Size: A3 | Date: 2026-10-31 | **Rev: P1** |
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| KiCad E.D.A. | kicad-cli 7.0.11+dfsg-1build4 | Id: 1/1 |